
LECTURE NOTES

Lecture 2a - Background: Linear Algebra & Probability

AI506 — Advanced Machine Learning

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Tensor

An array of numbers (like vectors or matrices) but with an arbitrary number of axes. The number of axes is denoted the **rank** of a tensor.

Example

Multiple images with colour codes:

$$T \in \mathbb{R}^{\text{batchsize} \times h \times w \times c}$$

Sum rule

$$P(X = x_i) = \sum_{j=1}^L P(X = x_i, Y = y_j)$$

$$P(X) = \sum_Y P(X, Y)$$

Product rule

$$P(X, Y) = P(Y|X)P(X)$$

Bayes' Theorem

$$\underbrace{P(Y|X)}_{\text{posterior}} = \frac{\overbrace{P(X|Y)}^{\text{likelihood}} \overbrace{P(Y)}^{\text{prior}}}{\underbrace{P(X)}_{\text{evidence}}}$$

Entropy

Degree of uncertainty.

Example: a fair die has high uncertainty (all events are equally likely); a biased die has lower uncertainty.